

Causality and Machine Learning: *Estimation and Causal Inference*

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Course website:

- <https://shorturl.at/efTNq>
- <https://johandh2o.github.io/causal-inference-summer-school>



Quick survey

Please raise your hand if you are familiar with, have used, heard about, or read about:

- ① Potential outcomes
- ② S-learner, T-learner, X-learner
- ③ Doubly robust estimation, DML, TMLE

How we will work today

- Please raise your hand or interrupt me at any time if something is unclear.
- The session is designed to be mid-level. If you are interested in introductory or advanced material, I am happy to point you to further resources.
- Some topics will be covered quickly as a panoramic overview. Please contact me if you would like to dive deeper.
- We will use small exercises and labs to keep the session interactive.
- For any extra question, suggestion, or concern, contact me at johanmd@uio.no.

1. Causality refresher

Causal inference

Causal inference is the methodological and scientific process of drawing conclusions about the strength of **effects** induced on some variables in a system (**outcomes**) by **interventions or actions** applied to other variables in the same system (**treatments / exposures**).

Causal vs. statistical inference

Causal inference is fundamentally distinct from **statistical inference**.

- **External actions** that break natural relations vs. **passive relations** as they are observed.
- **Changes *ceteris paribus*** vs. **regular empirical changes** (all move, no holding constant).

Let A be a **treatment** and Y be an **outcome**:

Causal questions

- What would happen to Y if we set $A = a$?
- Would Y have changed if A had been different?
- Which intervention value a for A increases Y the most?

Statistical questions

- How are A and Y associated?
- How well does A predict Y ?
- What is the best model to predict Y using A ?
- How does Y vary across observed groups X ?

2. Formalisms

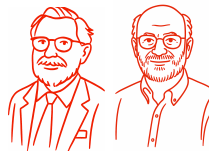
Formalisms

Pearl's Structural Causal Model (SCM)



- Primitive: an SCM or a causal graph,
- Identification: graphical conditions (+ positivity),
- Focus: hierarchy (associations $\rightarrow$ interventions $\rightarrow$ counterfactuals),
- Common users: engineers, computer scientists, psychologists...

Neyman–Rubin Causal Model (RCM)



- Primitive: potential outcomes,
- Identification: consistency + positivity + SUTVA + ignorability/exchangeability,
- Focus: randomized experiments for interventions, pseudo-exp, emulations,
- Common users: economists, epidemiologists, biostatisticians...

Potential outcomes

We have data on people in a population P and their treatment use:

- $W \in \mathcal{W}$ is **pre-exposure background** information, e.g. age, gender,...
- $A \in \{0, 1\}$ is a **binary point exposure**, e.g. 1=medication, 0=control,
- $Y \in \mathcal{Y}$ is a **continuous or discrete outcome**, e.g. clinical state, survival, progression....

Outcomes:

- Y_i is the **observed outcome** of unit i (under their actual treatment),
- Y_i^a is the **potential outcome** of unit i had they received treatment $A = a$,
 - Y_i^1 their potential outcome under medication,
 - Y_i^0 their potential outcome under control.

The fundamental problem of causal inference: for every unit i we only observe **one** of the potential outcomes, either if $A_i = 0$ then $Y_i^0 = Y_i$, or if $A_i = 1$ then $Y_i^1 = Y_i$; **never both**.

Consistency: If we see $A_i = a$ then $Y_i^a = Y_i$.

Potential outcomes, notation and SCMs

Many equivalent notations for potential outcomes

$$Y_i^a, \quad Y_i^{A=a}, \quad Y_i^{A \leftarrow a}, \quad Y_i(a), \quad Y^a(u_i).$$

Potential outcomes are central in the RCM, but can be derived from an SCM too.

If $Y = f_Y(W, A, u_Y)$ is the mechanism determining Y in an SCM, then $Y_i^a = Y^a(u_i)$ is the value that Y takes when A is replaced by a in its structural equation for a unit with identity $u_i = (u_{W,i}, u_{A,i}, u_{Y,i})$ (Pearl 2009).

$$Y_i^a = f_Y(W_i^a, a, u_{Y,i}),$$

where $W_i^a = W_i$ if W is pre-treatment.

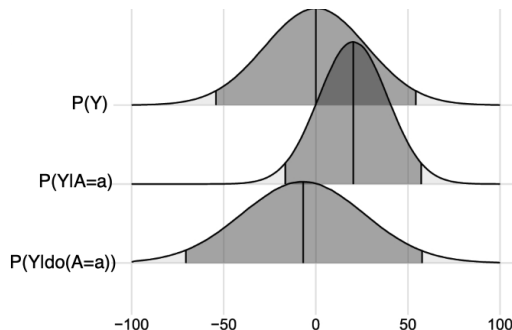
3. Formulation of causal effects, queries and estimands

Causal effects, queries and estimands

(1) We can ask: what is the distribution of an outcome **after** an intervention that sets $A = a$

The (post-)interventional distribution

$$P_Y(y \mid \text{do}(A = a)) \equiv P_{Y^a}(y)$$



Causal effects, queries and estimands

(2) We can ask: **what is the expected outcome after** an intervention that sets $A = a$

The expected potential outcome

$$\mathbb{E}[Y \mid \text{do}(A = a)] \equiv \mathbb{E}[Y^a]$$

(3) We can ask: **what is the expected outcome after** an intervention that sets $A = a$ for a subpopulation with $X = x$

The expected conditional potential outcome

$$\mathbb{E}[Y \mid \text{do}(A = a), X = x] \equiv \mathbb{E}[Y^a \mid X = x]$$

Why is $X = x$ not inside a *do*-operator?

Causal effects, queries and estimands

(4) We can ask: what is the **expected difference in outcomes between** two treatments $A = 1$ vs $A = 0$

Average treatment effect (ATE or ACE)

$$\text{ATE} = \mathbb{E}[Y \mid \text{do}(A = 1)] - \mathbb{E}[Y \mid \text{do}(A = 0)] \equiv \mathbb{E}[Y^1 - Y^0]$$

(5) We can ask: what is the **expected difference in outcomes between** two treatments $A = 1$ vs $A = 0$ for a subpopulation with $X = x$

Conditional ATE (CATE) - Heterogeneous treatment effect (HTE)

$$\text{CATE}(x) = \mathbb{E}[Y \mid \text{do}(A = 1), X = x] - \mathbb{E}[Y \mid \text{do}(A = 0), X = x] \equiv \mathbb{E}[Y^1 - Y^0 \mid X = x]$$

Two caveats to keep in mind about the CATE

Conditional average treatment effect

[1] While the ATE is typically lower-dimensional (a number or Euclidean vector), the CATE is typically higher-dimensional, especially if X is continuous. **It is a function.**

[2] X has to be **pre-treatment** for $\text{CATE}(x) = \mathbb{E}[Y^1 - Y^0 \mid X = x]$ to make sense. In other words, **there cannot be directed causal paths** $A \rightarrow X$

Why?

ATT and ATC: effects for policy-relevant subpopulations

The ATE averages the causal effect over the **full population**. Sometimes, however, the relevant target population is not everyone.

$$ATT = \mathbb{E}[Y^1 - Y^0 \mid A = 1]$$

Among those who actually received treatment, what was the average effect of receiving it?

$$ATC = \mathbb{E}[Y^1 - Y^0 \mid A = 0]$$

Among those who did not receive treatment, what would be the average effect of receiving it?

These estimands are useful when treatment effects are **heterogeneous** and the scientific or policy question targets a specific group: the currently treated or the currently untreated.

ATE = ATT when

- treatment effects are constant: $Y^1 - Y^0 = \tau$ for everyone;
- treatment assignment is independent of the individual treatment effect: $A \perp\!\!\!\perp (Y^1 - Y^0)$.

Marimo lab

- <https://shorturl.at/h02wM>
- <https://johandh2o.github.io/causal-inference-summer-school/day3.html>



You can interact online (Chrome is better), or download files and run locally. Required packages

```
pip3 install numpy pandas scipy scikit-learn matplotlib statsmodels  
jupyterlab marimo pgmpy
```

```
marimo edit /your-path/your-folder
```

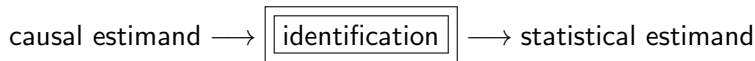
4. The causal roadmap

The causal roadmap

5. Identification from observational data

Identification

Is it possible to express the causal effect as a **statistical** estimand?
 Is it possible to learn the causal effect from **observational / passive data**?



Both frameworks (SCM/RCM) use **graphical criteria** for identification, but the RCM focuses more on **their consequences in terms of conditional independence using potential outcomes**.

Let W be a set of pre-treatment variables:

$$Y \perp\!\!\!\perp_d A \mid W \text{ in graph } \mathcal{G}_{\underline{A}}$$

A **d -separation condition** on $\mathcal{G}_{\underline{A}}$ (deleting the arrows out of A), or an application of **do-calculus**.

$$Y^a \perp\!\!\!\perp A \mid W \text{ for all } a \in \text{dom } A$$

Ignorability/exchangeability condition:
 given W , treatment assignment is as-good-as random for the potential outcomes.

Example: ATE and the backdoor criterion

Suppose we observe data $O = (W, A, Y) \sim P$:

- $W \in \mathcal{W}$ is **pre-exposure background** information, e.g. age, gender,...
- $A \in \{0, 1\}$ is a **binary point exposure**, e.g. 1=medication, 0=control,
- $Y \in \mathcal{Y}$ is a **continuous or discrete outcome**, e.g. clinical state, survival, progression...

We want to identify the $\text{ATE} = \mathbb{E}[Y^1 - Y^0]$

We have the causal graph: $Y \leftarrow W \longrightarrow A \longrightarrow Y$ and the mutilated one removes $A \longrightarrow Y$

W satisfies the **backdoor criterion**, so we have $Y \perp\!\!\!\perp_d A \mid W$ in $\mathcal{G}_{\underline{A}} \Rightarrow Y^a \perp\!\!\!\perp A \mid W$

$$\mathbb{E}[Y^a] = \mathbb{E}_W \mathbb{E}[Y^a \mid W] \stackrel{\text{ign}}{=} \mathbb{E}_W \mathbb{E}[Y^a \mid W, A] \stackrel{\text{pos}}{=} \mathbb{E}_W \mathbb{E}[Y^a \mid W, A = a] \stackrel{\text{con}}{=} \mathbb{E}_W \mathbb{E}[Y \mid W, A = a]$$

$$\text{ATE} = \mathbb{E}[Y^1 - Y^0] = \mathbb{E}_W \mathbb{E}[Y \mid W, A = 1] - \mathbb{E}_W \mathbb{E}[Y \mid W, A = 0]$$

Let's pause and recap

- For two or more values of the treatment (medication vs control), **we never observe both for the same unit** (*the fundamental problem of C.I.*).
- Two formalisms: **RCM** [primitive: potential outcomes] (Neyman 1923; Rubin 1974) and **SCM** [primitive: causal graphs] (Pearl 2009).
- **Potential outcomes** Y_i^a are **unit-level** counterfactual values of the outcome Y under different hypothetical intervention scenarios for the exposure A .
- We can formulate different **causal effects, queries or estimands**. They are notationally expressed using potential outcomes Y^a or the **do-operator**.
- A common estimand is the **ATE**. It is typically a number/vector. Another common estimand is the **CATE**. It is typically a **function**.
- Before estimating a causal effect, we must first ensure that it is **well formulated** and **identified** under explicit assumptions.

Questions?

6. Estimation

Plug-in estimation

We have data of size n , $\text{data} = \{O_i\}_{i=1}^n$ with $O_i = (W_i, A_i, Y_i) \stackrel{iid}{\sim} P$. Let us denote $\psi = \text{ATE}$

$$\psi = \mathbb{E}_W \underbrace{\mathbb{E}[Y \mid W, A = 1]}_{\text{regression}} - \mathbb{E}_W \underbrace{\mathbb{E}[Y \mid W, A = 0]}_{\text{regression}}$$

The expectation outside can be approximated with the average, so an estimator could be:

$$\hat{\psi} = \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}[Y \mid W_i, A = 1] - \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}[Y \mid W_i, A = 0]$$

How should we fit such a regression model? Linear regression? **ML?**

Plug-in (ATE/CATE)

- ① S-learner
- ② T-learner
- ③ (X-learner)
- ④ NN architectures

Orthogonal and DR (ATE)

- ① AIPW
- ② One-step / DML
- ③ TMLE*

Heterogeneous (CATE)

- ① R-learner
- ② DR-learner
- ③ Causal forests**

Why are fully parametric models risky in observational causal inference?

linear outcome regression: $Y_i = \beta_0 + \beta_1 A_i + \beta_2^\top W_i + \epsilon_i$, with $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$.

In observational studies, causal identification already depends on strong assumptions. Adding rigid parametric models adds more assumptions and can produce **misspecification bias** and **poor inference**.

misspecification $\Rightarrow$ biased causal effect

- Outcome and treatment assignment are rarely simple.
- Missing interactions or nonlinearities.
- Chosen functions are often unrealistic.
- Selection on high dimensional data is challenging.
- **Identification is nonparametric!**
Estimation should be as well.

overconfidence $\Rightarrow$ unreliable inference

- Some strata have little or no overlap: parametric models fill gaps by extrapolating.
- Standard errors are too small, as they may ignore sources of uncertainty.
- Confidence intervals may have undercoverage, **with coverage tending to zero.**

Rather than relying only on rigid parametric models, modern causal inference prefers nonparametric models and ML

Plug-in estimation: S-learner

Based on a **single** outcome regression surface. Recipe:

- 1 Learn one regression on the whole data using a loss function L and a model class $\mathcal{Q}$:

$$\hat{Q} = \arg \min_{Q \in \mathcal{Q}} \frac{1}{n} \sum_{i=1}^n L(Y_i, Q(W_i, A_i)).$$

- 2 For each unit i , predict the outcome under treatment and under control:

two predictions for i : $\hat{Q}(W_i, 1)$ and $\hat{Q}(W_i, 0)$.

- 3 Compute the difference, and average:

$$\hat{\psi}_S = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{Q}(W_i, 1) - \hat{Q}(W_i, 0) \right\}.$$

Summary: learn one outcome surface and compare its predictions when A is set to 1 versus 0.

Plug-in estimation: T-learner

Based on **two** regression surfaces. Recipe:

- 1 Learn one outcome regression **for each treatment arm** using loss L and model class $\mathcal{Q}$:

$$\hat{Q}_1 = \arg \min_{Q \in \mathcal{Q}} \frac{1}{n_1} \sum_{i:A_i=1} L(Y_i, Q(W_i)),$$

$$\hat{Q}_0 = \arg \min_{Q \in \mathcal{Q}} \frac{1}{n_0} \sum_{i:A_i=0} L(Y_i, Q(W_i)).$$

- 2 For each unit i , predict with both fitted models:

two predictions for i : $\hat{Q}_1(W_i)$ and $\hat{Q}_0(W_i)$.

- 3 Compute the difference, and average:

$$\hat{\psi}_T = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{Q}_1(W_i) - \hat{Q}_0(W_i) \right\}.$$

Summary: learn one outcome surface among treated units and another among controls, then contrast them.

S-learner vs T-learner

S-learner: **advantages**

- Uses all observations to learn one outcome model.
- Simple to implement with any ML algorithm.
- Can be stable when sample sizes are small.

S-learner: **disadvantages**

- Treats A like any other “feature”, and may underuse it.
- May shrink treatment effects toward zero.
- Can ignore effect heterogeneity (CATE).

T-learner: **advantages**

- Allows different outcome models by treatment arm.
- More flexible for heterogeneous effects.
- Easy to interpret: treated model vs control model.

T-learner: **disadvantages**

- Each model uses only part of the data.
- Can be unstable with imbalanced treatment groups.
- May extrapolate poorly where one arm has little support.

Marimo lab

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Lunch time!

But... there are some caveats about the use of data-adaptive ML methods in causal inference

Why is flexible ML plug-in risky in causal inference?

ML outcome plug-in: $\hat{Q} = \arg \min_{Q \in \mathcal{Q}} \frac{1}{n} \sum_{i=1}^n (Y_i - Q(W_i, A_i))^2$: min MSE, best BV trade-off.

Flexible ML helps reduce misspecification, but most ML algorithms are designed for **prediction**, not directly for **causal estimation** or **valid inference**.

prediction loss $\nrightarrow$ causal accuracy

- Optimizes prediction of Y , not bias of $\hat{\psi}$.
- Bias-variance trade-offs may accept too much bias for the target parameter.
- **It only uses a model for Q , which ignores other components such as the treatment model..**

black-box fitting $\Rightarrow$ hard inference

- Many ML estimators have complex asymptotics.
- Standard errors for $\hat{\psi}$ are not automatic.
- Overfitting can invalidate inference.
- Uncertainty in nuisance functions is hard to propagate.

Let's pause and recap

- Causal estimation should only begin once the target effect is **well formulated** and **identified** under explicit assumptions.
- Parametric models are simple and interpretable, but can be highly biased when their functional form is **misspecified**.
- Machine learning offers flexible plug-in estimation strategies, such as S-, T-, and X-learners.
- These learners involve trade-offs: S-learners may smooth away heterogeneous effects, while T-learners can suffer from instability under small samples or weak overlap.
- The central warning is that prediction and causal inference are not the same: a model can predict well and still estimate causal effects poorly and might not provide valid inference.

Questions?

7. Orthogonal estimation with semiparametric theory

Inverse probability weighting (IPW)

A key equality via **inverse probability weighting (IPW)** or **importance sampling** is:

$$\mathbb{E}_W \mathbb{E}[Y \mid W, A = a] = \mathbb{E} \left[\frac{\mathbb{I}(A = a) Y}{\mathbb{P}(A = a \mid W)} \right],$$

which is valid under **positivity** $\mathbb{P}(A = a \mid W = w) > 0$ for all w .

Proof.

$$\begin{aligned} \mathbb{E}_W \mathbb{E}[Y \mid W, A = a] &= \int p(w) \int y p(y \mid w, a) dy dw \\ &= \int p(w) \sum_{a' \in \{0,1\}} \int \mathbb{I}(a' = a) y p(y \mid w, a') dy dw \\ &= \int p(w) \sum_{a' \in \{0,1\}} \int \frac{\mathbb{I}(a' = a)}{p(a' \mid w)} y p(y, a' \mid w) dy dw \\ &= \mathbb{E}_W \mathbb{E}_{A,Y \mid W} \left[\frac{\mathbb{I}(A = a) Y}{p(a \mid W)} \right] = \mathbb{E} \left[\frac{\mathbb{I}(A = a) Y}{\mathbb{P}(A = a \mid W)} \right]. \end{aligned}$$

AIPW estimation

Plug-in estimators only use a model for the outcome regression Q . Can we leverage a model for A , $\pi(a | w) = \mathbb{P}(A = a | W = w)$ (**propensity score**) as well?

$$\begin{aligned}
 \mathbb{E}[Y^a] &= \mathbb{E}_W \mathbb{E}[Y | W, A = a] && \text{identification} \\
 &= \mathbb{E}_W \mathbb{E}_{A, Y | W} \left[\frac{\mathbb{I}(A = a) Y}{\pi(a | W)} \right] && \text{IPW} \\
 &= \mathbb{E}_W \mathbb{E}_{A, Y | W} \left[\frac{\mathbb{I}(A = a) (Y - Q(W, A) + Q(W, A))}{\pi(a | W)} \right] && \text{add/subtract } Q \\
 &= \mathbb{E}_W \mathbb{E}_{A, Y | W} \left[\frac{\mathbb{I}(A = a) (Y - Q(W, A))}{\pi(a | W)} + \frac{\mathbb{I}(A = a) Q(W, a)}{\pi(a | W)} \right] && \text{distribute} \\
 &= \underbrace{\mathbb{E}_W \mathbb{E}_{A, Y | W} \left[\frac{\mathbb{I}(A = a) (Y - Q(W, A))}{\pi(a | W)} \right]}_{\text{one-step correction}} + \underbrace{\mathbb{E}_W Q(W, a)}_{\text{plug-in}} && \text{IPW}
 \end{aligned}$$

This is the **Augmented Inverse-Probability Weighting (AIPW)** technique (Robins et al. 1994).

The double robustness property

ATE estimators:

$$\hat{\psi}_{\text{plug}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{Q}(W_i, 1) - \hat{Q}(W_i, 0) \right],$$

$$\hat{\psi}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{Q}(W_i, 1) - \hat{Q}(W_i, 0) + \left\{ \frac{A_i}{\hat{\pi}(1 | W_i)} - \frac{1 - A_i}{\hat{\pi}(0 | W_i)} \right\} \{Y_i - \hat{Q}(W_i, A_i)\} \right].$$

The plug-in estimator depends only on the outcome model for Q . If $\hat{Q}$ is misspecified, then $\hat{\psi}_{\text{plug}}$ is generally biased.

The AIPW estimator combines an outcome model $\hat{Q}$ and a propensity model $\hat{\pi}$. It is **doubly robust** (Bang and Robins 2005): it is consistent if either $\hat{Q}$ or $\hat{\pi}$ is consistently estimated.

Two chances to remove bias: outcome regression correction + inverse-probability residual correction. DR is one of the most impactful properties in modern causal inference! It is like Actor-Critic in RL.

The influence function

The AIPW estimator is a special case of a broader idea: **a one-step corrected estimator using a first-order expansion** (von Mises 1947).

Ordinary Taylor expansion

$$f(\hat{x}) \approx f(x) + (\hat{x} - x) f'(x).$$

Statistical von Mises expansion

$$\psi(\hat{P}) \approx \psi(P) + (\hat{\mathbb{E}} - \mathbb{E}) D_P(O).$$

Here D_P is the **influence function**: the first-order derivative of the causal estimand with respect to changes in the distribution of the data $O = (W, A, Y)$.

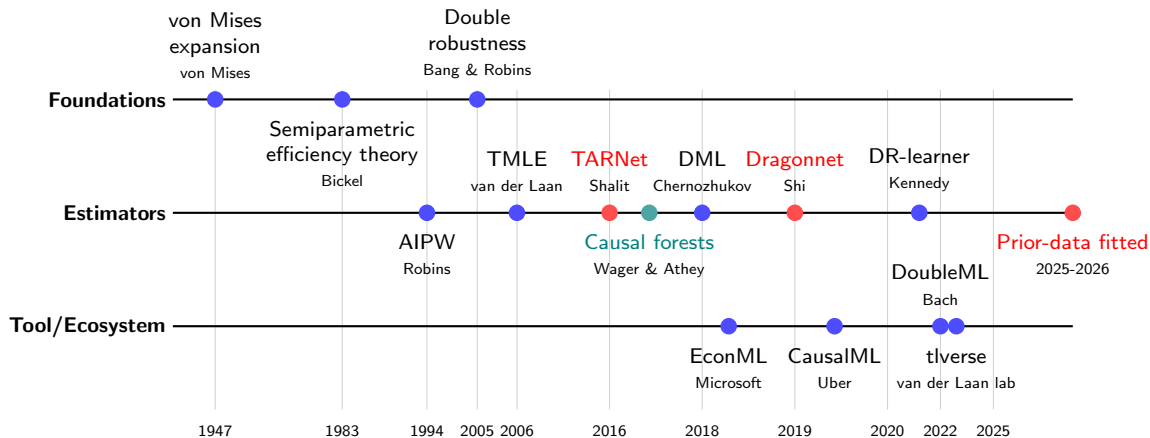
For the ATE it gives:

$$D_P(O) = \underbrace{\left\{ \frac{A}{\pi(1|W)} - \frac{1-A}{\pi(0|W)} \right\} \{Y - Q(W, A)\}}_{\text{one-step/first-order correction}} + \underbrace{Q(W, 1) - Q(W, 0)}_{\text{plug-in}} - \underbrace{\psi}_{\text{ATE}}$$

the score $\varphi(O)$

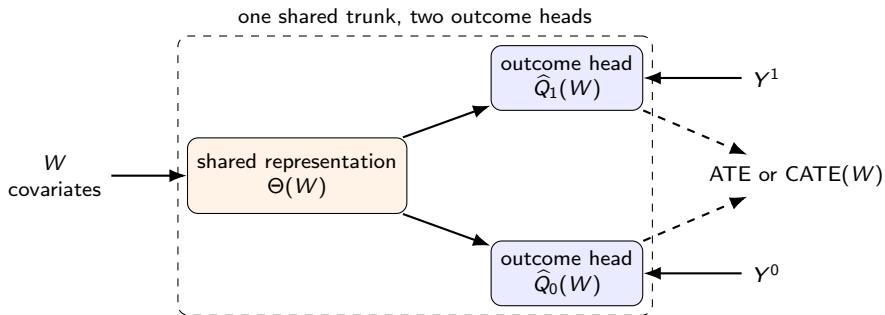
Property: $\mathbb{E}D_P(O) = 0$. Move ψ to the other side and one gets the AIPW!

A schematic timeline of semiparametric theory + causal ML



Methods based on **semiparametric theory** (influence functions) are dominant in today's causal ML, while **emerging directions using neural networks and foundation models are on the rise**.

Meta-learning with neural nets: TARNet



Features (Shalit et al. 2017)

- Learns a shared representation $\Theta(W)$.
- Separate outcome heads: treated and control.
- Closer to a T-learner.

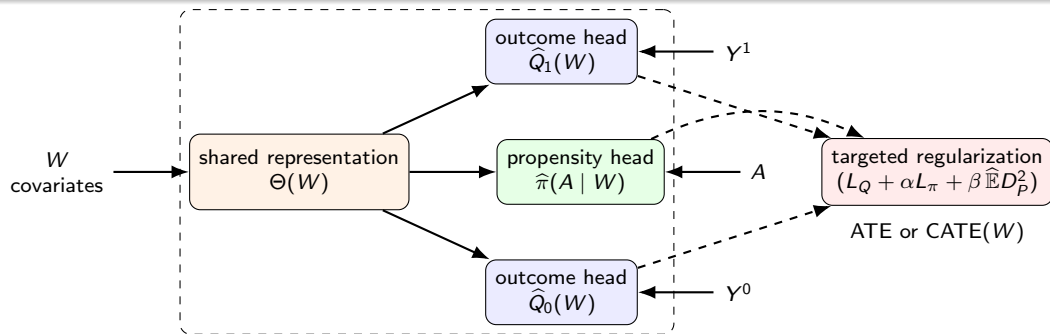
Advantages

- Shares information across arms through the common trunk (like an S-learner).
- Often more stable than fitting two unrelated deep nets.

Disadvantages

- Prediction-oriented.
- Does not explicitly model the propensity score.
- **No DR, no automatic inference.**
- Biased under poor overlap.

Meta-learning with neural nets: Dragonnet



Features (Shi et al. 2019)

- TARNet + propensity head.
- Learns $\hat{Q}_1(w)$, $\hat{Q}_0(w)$, and $\hat{\pi}(a | w)$ jointly.
- Includes targeted regularization inspired by semiparametric ideas.

Advantages

- Better aligned with causal adjustment than pure prediction nets.
- Uses both outcome and treatment-assignment information.

Disadvantages

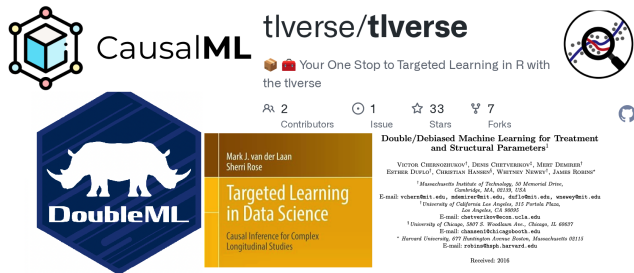
- Complex to tune / train.
- **No DR no inference.**
- Harder to interpret than simpler semiparametric estimators.

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State of the art: orthogonal estimation



CausalML

tlverse/tlverse
Your One Stop to Targeted Learning in R with the tlverse

DoubleML

Mark J. van der Laan
Sherri Rose

Targeted Learning in Data Science
Causal Inference for Complex Longitudinal Studies

Double/Debiased Machine Learning for Treatment and Structural Parameters¹

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Received: 2016

Orthogonal estimators are now central

- Combine flexible ML with semiparametric theory.
- Often provide double robustness.
- Valid asymptotic inference after ML nuisance estimation.
- Have become highly visible in statistics, econometrics, ML, and AI venues.

But there are important challenges

- Mathematically advanced, with a steep learning curve.
- Require careful implementation: cross-fitting, nuisance estimation, positivity checks.
- Less plug-and-play than standard predictive ML workflows.

DML and one-step estimation

DML is essentially **AIPW / one-step estimation** with two extra ingredients: **flexible ML** for nuisance functions and **cross-fitting** to avoid using the same data twice (Chernozhukov et al. 2018).

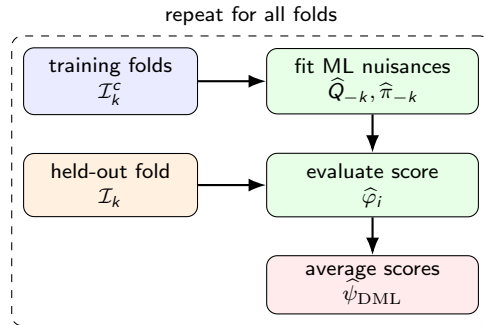
$$Q(w, a) = \mathbb{E}[Y \mid W = w, A = a], \quad \pi(a \mid w) = \mathbb{P}(A = a \mid W = w).$$

For unit i in fold $k(i)$, estimate them on the other folds: $\hat{\eta}_{-k(i)} = (\hat{Q}_{-k(i)}, \hat{\pi}_{-k(i)})$.

Then evaluate the AIPW score out-of-sample:

$$\begin{aligned} \hat{\varphi}_i &= \hat{Q}_{-k(i)}(W_i, 1) - \hat{Q}_{-k(i)}(W_i, 0) \\ &+ \left\{ \frac{A_i}{\hat{\pi}_{-k(i)}(1 \mid W_i)} - \frac{1 - A_i}{\hat{\pi}_{-k(i)}(0 \mid W_i)} \right\} \\ &\times \{Y_i - \hat{Q}_{-k(i)}(W_i, A_i)\}. \end{aligned}$$

$$\hat{\psi}_{\text{DML}} = \frac{1}{n} \sum_{i=1}^n \hat{\varphi}_i$$



Why is DML debiased?

Reorder terms in the expansion:

$$\hat{\psi}_{\text{DML}} - \psi = \underbrace{(\hat{\mathbb{E}} - \mathbb{E})D_P(O)}_{\text{sampling noise}} + \underbrace{R_1(\hat{P}, P)}_{\text{1-order bias}} + \underbrace{R_2(\hat{P}, P)}_{\text{2-order remainder}} + o_P(n^{-1/2})$$

[1] The orthogonal scores in DML **force the 1-order bias to be zero**

$$R_1(\hat{P}, P) := \frac{d}{dt} \mathbb{E} \left[\underbrace{\psi(P_t) + D_{P_t}(O)}_{\text{the score } \varphi_t(O)} \right]_{t=0} = 0, \quad P_t = P + t(\hat{P} - P).$$

[2] If models converge fast enough, $\|\hat{Q} - Q\| = o_P(n^{-1/4})$ and $\|\hat{\pi} - \pi\| = o_P(n^{-1/4})$, one has

$$|R_2(\hat{P}, P)| \lesssim \|\hat{Q} - Q\| \|\hat{\pi} - \pi\| = o_P(n^{-1/2}).$$

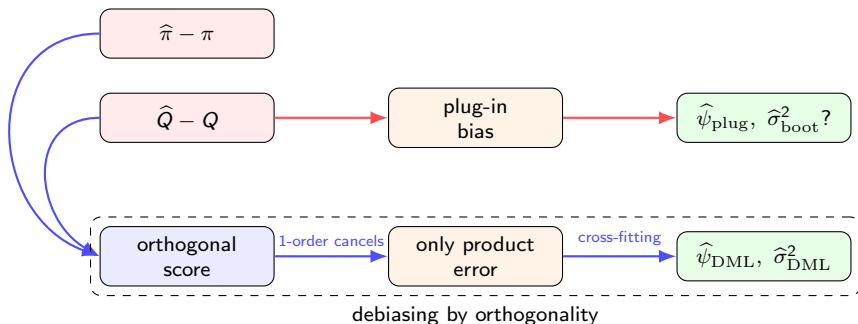
[3] Cross-fitting controls the empirical fluctuation from using $D_{\hat{P}}$:

$$(\hat{\mathbb{E}} - \mathbb{E})D_{\hat{P}}(O) = (\hat{\mathbb{E}} - \mathbb{E})D_P(O) + o_P(n^{-1/2}).$$

We get an asymptotic normal distribution

$$\hat{\psi}_{\text{DML}} - \psi = (\hat{\mathbb{E}} - \mathbb{E})D_{\hat{P}}(O) + o_P(n^{-1/2}) \Rightarrow \hat{\psi}_{\text{DML}} \approx N\left(\psi, \sigma_{\text{DML}}^2 = n^{-1} \mathbb{E} D_{\hat{P}}^2(O)\right)$$

Why is DML debiased?



DML has become popular because it combines several attractive properties:

- it allows the use of flexible machine learning methods;
- it can deliver double robustness;
- it provides a principled route to confidence intervals.

But... suppose we are estimating **a probability**. The plug-in estimate is 0.95, but the one-step correction moves it by 0.06. What should the final estimate be?

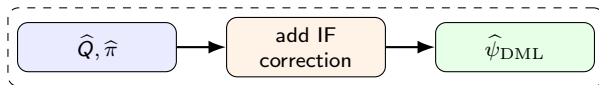
Targeted Minimum-Loss Estimation (TMLE)

TMLE starts with flexible ML, but then **targets** the initial regression toward the estimand ψ (van der Laan and Rose 2011, 2018).

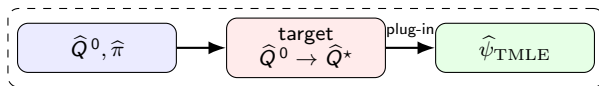
$$\hat{\psi}_{\text{plug}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{Q}^0(W_i, 1) - \hat{Q}^0(W_i, 0) \right].$$

$$\hat{\psi}_{\text{TMLE}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{Q}^*(W_i, 1) - \hat{Q}^*(W_i, 0) \right].$$

DML / one-step



targeted plug-in



TMLE respects the natural bounds of the parameters, but it is **asympt. equivalent to DML**.

TMLE: the targeting step

Remember the influence function

$$D_P(O) = \underbrace{\left\{ \frac{A}{\pi(1|W)} - \frac{1-A}{\pi(0|W)} \right\}}_{\text{one-step/first-order correction}} \underbrace{\{Y - Q(W, A)\}}_{\text{the score } \varphi(O)} + \underbrace{Q(W, 1) - Q(W, 0)}_{\text{plug-in}} - \underbrace{\psi}_{\text{ATE}}$$

The red part is called the **clever covariate** or **Riesz representer** of the ATE:

$$H(W, A) = \frac{A}{\pi(1|W)} - \frac{1-A}{\pi(0|W)}.$$

The **targeting step** updates the **initial outcome regression** $\hat{Q}^0$ in the direction of the clever covariate. **With an L^2 -loss, this is the one-dimensional least-squares fluctuation:**

$$\hat{\epsilon} = \frac{\sum_{i=1}^n H_{\hat{\pi}}(W_i, A_i) \{Y_i - \hat{Q}^0(W_i, A_i)\}}{\sum_{i=1}^n H_{\hat{\pi}}(W_i, A_i)^2}$$

OLS coeff.: residuals $\sim H$ with no intercept,

$$\hat{Q}^*(W, A) = \hat{Q}^0(W, A) + \hat{\epsilon} H_{\hat{\pi}}(W, A).$$

This makes the one-step correction approx. zero. Other loss functions produce other updates.

Let's pause and recap

- Blindly using ML for causal estimation can produce biased plug-in estimates: good prediction does not guarantee valid causal inference. This is why **debiased** and **double-robust** estimators are desirable.
- AIPW combines an outcome model $\hat{Q}$ and a treatment model $\hat{\pi}$, giving **double robustness**: consistency can hold if either nuisance model is correct.
- DML and one-step estimators use ML for nuisance functions, but rely on **orthogonality** and **cross-fitting** to control overfitting and nuisance bias.
- TMLE also uses the influence function, but instead of only correcting $\hat{\psi}$, it **targets** $\hat{Q}$ so the final plug-in estimate respects the parameter space.

Questions?

Marimo lab

- <https://shorturl.at/h02wM>
- <https://johandh2o.github.io/causal-inference-summer-school/day3.html>



8. Heterogeneous treatment effect estimation

Heterogeneous treatment effects

We keep the same observational data structure:

$$O = (W, A, Y) \sim P, \quad A \in \{0, 1\}, \quad W \text{ pre-treatment.}$$

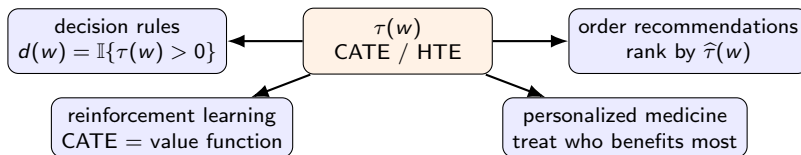
$$\text{ATE} = \psi := \mathbb{E}[Y^1 - Y^0] = \mathbb{E}_W[\tau(W)],$$

$$\text{CATE} = \tau(w) := \mathbb{E}[Y^1 - Y^0 \mid W = w]$$

Ignorability + Positivity: $\tau(w) = Q(w, 1) - Q(w, 0)$, $Q(w, a) := \mathbb{E}[Y \mid W = w, A = a]$.

The ATE answers “does treatment work on average?” whereas the CATE asks **for whom and under which context does treatment work?** (Künzel et al. 2019; Wager and Athey 2018).

It is a very important object for decision-making.



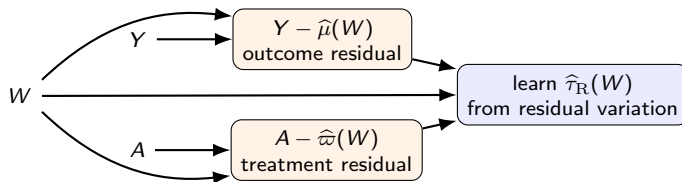
R-learner: remove the parts that are easy to predict

Define two nuisance functions:

$$\mu(w) := \mathbb{E}[Y \mid W = w], \quad \varpi(w) := \mathbb{P}(A = 1 \mid W = w) \equiv \pi(1 \mid w).$$

The key **Robinson residualization** (Robinson 1988) is

$$Y - \mu(W) = \{A - \varpi(W)\} \tau(W) + \varepsilon, \quad \mathbb{E}[\{A - \varpi(W)\} \varepsilon \mid W] = 0.$$



Intuition: first remove the part of Y and A explained by W ; what remains is the causal contrast signal (Nie and Wager 2021).

R-learner: recipe

A practical recipe:

- ① Split the data or use cross-fitting.
- ② Estimate nuisance functions with flexible ML:

$$\hat{\mu}(w) \approx \mathbb{E}[Y \mid W = w], \quad \hat{\varpi}(w) \approx \mathbb{P}(A = 1 \mid W = w).$$

- ③ Construct the residualized pseudo-regression problem:

$$\tilde{Y}_i := \frac{Y_i - \hat{\mu}(W_i)}{A_i - \hat{\varpi}(W_i)}, \quad \hat{\omega}_i := \{A_i - \hat{\varpi}(W_i)\}^2.$$

- ④ Fit any weighted ML regression **of the pseudo-outcome $\tilde{Y}$ against W (not including the treatment A)**.

$$\hat{\tau}_R \in \arg \min_{\tau \in \mathcal{T}} \frac{1}{n} \sum_{i=1}^n \hat{\omega}_i \{\tilde{Y}_i - \tau(W_i)\}^2.$$

Why useful?

- Learns $\tau(w)$ directly.
- Orthogonal to first-order nuisance error.
- Can use forests, splines, boosting, NN...

What to watch?

- **Not doubly robust in the AIPW sense.**
- Poor overlap creates instability.

DR-learner: turn CATE estimation into regression

The DR-learner builds a **doubly robust pseudo-outcome** for the CATE. Estimate

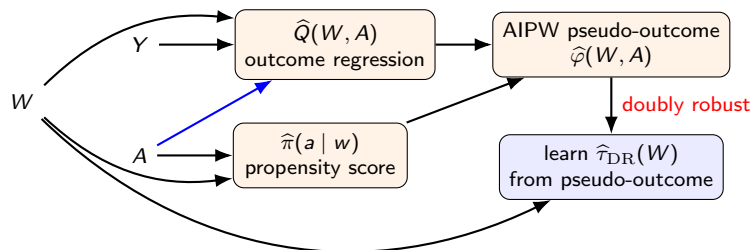
$$Q(w, a) = \mathbb{E}[Y \mid W = w, A = a], \quad \pi(a \mid w) = \mathbb{P}(A = a \mid W = w).$$

For each observation, use the **AIPW / DML score** as the **pseudo-outcome**:

$$\hat{\varphi}_i := \hat{Q}(W_i, 1) - \hat{Q}(W_i, 0) + \left\{ \frac{A_i}{\hat{\pi}(1 \mid W_i)} - \frac{1 - A_i}{\hat{\pi}(0 \mid W_i)} \right\} \{Y_i - \hat{Q}(W_i, A_i)\}.$$

Then estimate the CATE by regressing this pseudo-outcome on W (Kennedy 2023):

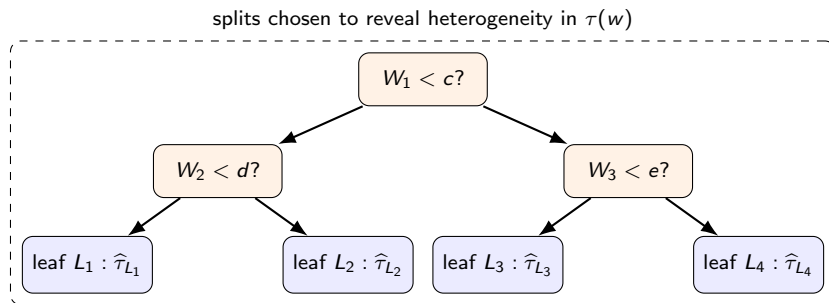
$$\hat{\tau}_{\text{DR}} \in \arg \min_{\tau \in \mathcal{T}} \frac{1}{n} \sum_{i=1}^n \{\hat{\varphi}_i - \tau(W_i)\}^2.$$



Causal forests: trees for treatment effects

A standard regression tree splits covariate space to improve prediction of Y .

A **causal tree / forest splits covariate space to find regions with different causal effects**.



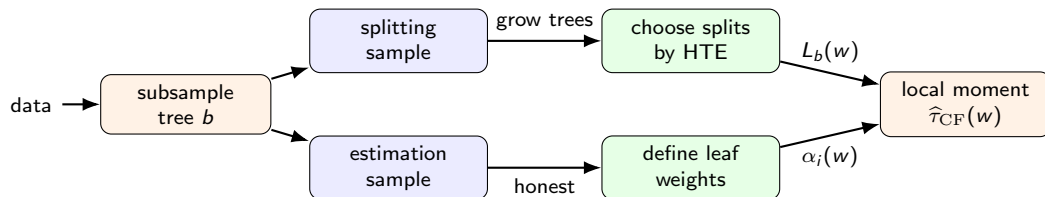
Inside a leaf L , in a randomized experiment one could estimate

$$\hat{\tau}_L = \frac{1}{n_{1,L}} \sum_{i: W_i \in L, A_i=1} Y_i - \frac{1}{n_{0,L}} \sum_{i: W_i \in L, A_i=0} Y_i.$$

In observational studies, causal forests use **sample splitting** and **orthogonal local moments** (Athey and Imbens 2016; Wager and Athey 2018)

Causal forests: learning algorithm

A causal forest learns many causal trees L_b . The weights $\alpha_i(w)$ are then induced by how often unit i falls in the same leaf as w .



Splits are chosen to create leaves with different causal effects. Given a target point w , let $L_b(w)$ be the leaf containing w in tree b . With honesty, weights are computed as:

$$\alpha_i(w) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{I}\{i \in \mathcal{N}_b^{\text{est}}(w)\}}{|\mathcal{N}_b^{\text{est}}(w)|}, \quad \mathcal{N}_b^{\text{est}}(w) := \{j \in \mathcal{I}_b^{\text{est}} : W_j \in L_b(w)\}.$$

These learned weights define a local R-learner:

$$\hat{\tau}_{CF}(w) = \frac{\sum_{i=1}^n \alpha_i(w) \{A_i - \hat{\omega}(W_i)\} \{Y_i - \hat{\mu}(W_i)\}}{\sum_{i=1}^n \alpha_i(w) \{A_i - \hat{\omega}(W_i)\}^2}.$$

Comparing HTE learners

All these methods estimate the same **function**,

$$\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w] \stackrel{id}{=} Q(w, 1) - Q(w, 0),$$

- **Plug-in:** $\hat{\tau}(w) = \hat{Q}(w, 1) - \hat{Q}(w, 0)$: Simple and modular, but mostly prediction-driven.
- **R-learner:** learn $\tau(w)$ from residual variation:

$$Y - \hat{\mu}(w) \approx \{A - \hat{\omega}(w)\}\tau(w).$$

Good for flexible ML, but not **really** doubly robust. Biased under poor overlap.

- **DR-learner:** regress a doubly robust pseudo-outcome $\hat{\varphi}$ on W . It can be unstable under **very poor overlap**. In practice, it can smooth some extreme pseudo-outcomes. Constructing reliable theory-based confidence regions for heterogeneous effects remains challenging.
- **Causal forests** learn adaptive neighborhoods around w and estimate local orthogonal effects. They are powerful for discovering heterogeneity, but more complicated and still not doubly robust.

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- <https://shorturl.at/h02wM>
- <https://johandh2o.github.io/causal-inference-summer-school/day3.html>



9. Final recap and conclusions

Final recap: ML for causal estimation

- **Causal estimation starts before ML.** Before estimating any effect, we need a well-defined causal estimand, identification assumptions, and a statistical estimand:
causal question $\longrightarrow$ identification $\longrightarrow$ estimation.
- **ML is useful because nuisance functions are hard.** Flexible models can estimate objects such as $Q(w, a)$, $\varpi(w)$, $\mu(w)$, and $\tau(w)$ without imposing strong parametric structure.
- **Prediction alone is not enough for causal inference.** Plug-in learners can be biased by nuisance misspecification, so orthogonality, debiasing, double robustness, and sample splitting are central tools.
- **HTE estimation moves from average effects to decisions.** Estimating $\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w]$ supports targeting, recommendations, personalized medicine, and policy learning, but requires stronger local overlap and more careful validation.
- **Many challenges remain.** Modern causal ML still faces difficult problems: continuous treatments, inference for continuous CATEs, missing data, selection bias, interference, and reliable uncertainty quantification with adaptive ML.

*Contact me if you want to make your final project about **estimation of causal effects with ML** (johanmd@uio.no). Thank you!*

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10. Extra slides

Other common causal estimands

- **Causal risk difference or ratio** for binary outcomes:

$$\text{RD} = \mathbb{P}(Y^1 = 1) - \mathbb{P}(Y^0 = 1),$$

$$\text{RR} = \frac{\mathbb{P}(Y^1 = 1)}{\mathbb{P}(Y^0 = 1)}.$$

- **Causal odds ratio** for binary outcomes:

$$\text{OR} = \frac{\mathbb{P}(Y^1 = 1)/\mathbb{P}(Y^1 = 0)}{\mathbb{P}(Y^0 = 1)/\mathbb{P}(Y^0 = 0)}.$$

- **Local average treatment effect** for compliers, using an instrument Z :

$$\text{LATE} = \mathbb{E}[Y^1 - Y^0 \mid A^1 > A^0].$$

- **Controlled direct effect** fixing a mediator $\text{do}(A = 1, M = m)$ vs $\text{do}(A = 0, M = m)$:

$$\text{CDE}(m) = \mathbb{E}[Y^{1,m} - Y^{0,m}].$$

- **Natural direct and indirect effects**:

$$\text{NDE} = \mathbb{E}[Y^{1,M^0} - Y^{0,M^0}],$$

$$\text{NIE} = \mathbb{E}[Y^{1,M^1} - Y^{1,M^0}].$$

Plug-in estimation: X-learner

Based on imputed individual treatment effects. Recipe:

- 1 First learn two outcome regressions, as in the T-learner:

$$\hat{Q}_1(w) = \hat{\mathbb{E}}[Y \mid W = w, A = 1], \quad \hat{Q}_0(w) = \hat{\mathbb{E}}[Y \mid W = w, A = 0].$$

- 2 Impute the missing treatment effect for each observed unit:

$$\hat{D}_i^1 = Y_i - \hat{Q}_0(W_i), \quad A_i = 1,$$

$$\hat{D}_i^0 = \hat{Q}_1(W_i) - Y_i, \quad A_i = 0.$$

- 3 Learn two treatment-effect regressions:

$$\hat{\tau}_1(w) = \hat{\mathbb{E}}[\hat{D}^1 \mid W = w, A = 1], \quad \hat{\tau}_0(w) = \hat{\mathbb{E}}[\hat{D}^0 \mid W = w, A = 0].$$

- 4 Combine them using the estimated propensity score $\hat{\varpi}(w) = \hat{\mathbb{P}}(A = 1 \mid W = w)$, and average:

$$\hat{\psi}_X = \frac{1}{n} \sum_{i=1}^n [\hat{\varpi}(W_i) \hat{\tau}_0(W_i) + \{1 - \hat{\varpi}(W_i)\} \hat{\tau}_1(W_i)].$$

Intuition: first impute the missing counterfactual outcomes, then learn and combine two treatment-effect models.

TMLE: targeted minimum-loss estimation

TMLE starts with flexible ML, but then **targets** the initial regression toward the estimand ψ .
Start with initial estimates

$$\hat{Q}^0(W, A), \quad \hat{\pi}(A | W).$$

Fluctuate the initial outcome regression, and regress **with no intercept** against the clever covariates

$$Y \sim \hat{Q}^0(W, A) + \epsilon H_{\hat{\pi}}(W, A).$$

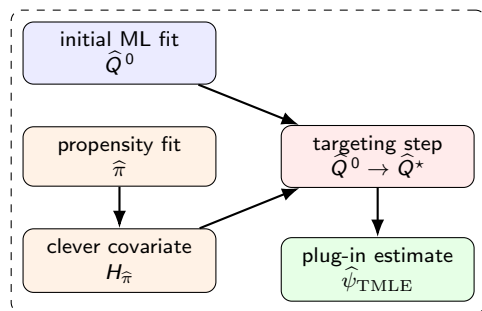
Estimate ϵ by OLS or by **minimizing a loss**

$$\hat{\epsilon} = \arg \min_{\epsilon} \frac{1}{n} \sum_{i=1}^n L \left\{ Y_i, \hat{Q}^{\epsilon}(W_i, A_i) \right\}.$$

Finally set **link** $\hat{Q}^* = \text{link } \hat{Q}^0 + \hat{\epsilon} H_{\hat{\pi}}$ and plug in:

$$\hat{\psi}_{\text{TMLE}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{Q}^*(W_i, 1) - \hat{Q}^*(W_i, 0) \right].$$

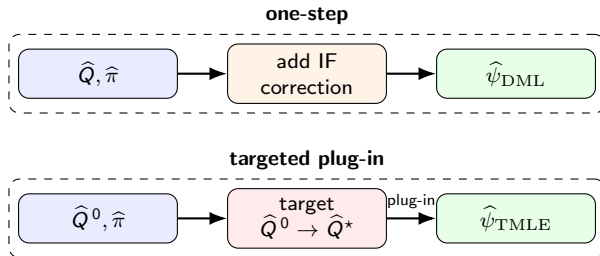
TMLE = ML + causal targeting



The update is usually low-dimensional. We do not refit the whole ML model. We only move $\hat{Q}^0$ in the direction that matters for the influence function.

What is the targeting step doing?

DML and TMLE use essentially the same influence-function idea, but implement it differently.



DML / one-step: add the correction after fitting

$$\hat{\psi}_{\text{DML}} = \hat{\mathbb{E}} \left[\hat{Q}(W, 1) - \hat{Q}(W, 0) \right] + \hat{\mathbb{E}} \left[H_{\hat{\pi}}(A, W) \{Y - \hat{Q}(W, A)\} \right].$$

TMLE: update the fit until the correction is zero

$$\hat{\psi}_{\text{TMLE}} = \hat{\mathbb{E}} \left[\hat{Q}^*(W, 1) - \hat{Q}^*(W, 0) \right] \text{ with } \hat{\mathbb{E}} \left[H_{\hat{\pi}}(A, W) \{Y - \hat{Q}^*(W, A)\} \right] \approx 0.$$

substitution estimator

respects bounds

asymptotically equivalent

DML corrects $\hat{\psi}$; TMLE corrects $\hat{Q}$